

Deepak Mallampati

Software Engineer - Generative AI & Full-Stack Systems

(330) 554-0770 | deepakmallampati00@gmail.com | [linkedin.com/in/deepak-mallampati](https://www.linkedin.com/in/deepak-mallampati) | US-based, F-1 OPT, open to sponsorship

SUMMARY

Software engineer with 4+ years building and shipping production Generative AI systems and full-stack products end-to-end across front-end, back-end, and infrastructure. Delivered 12 APIs and microservices serving 100,000+ requests daily across 20+ teams, cut p95 latency from 1.5s to 800ms, and shipped MangaLens - a live Chrome Web Store product running across 29+ sites. Fluent in Python, TypeScript/JavaScript, React, and FastAPI, with MongoDB and semantic/vector search in production - ready to build high-impact GenAI dataset systems at scale.

CORE COMPETENCIES

Full-Stack Development • Python & FastAPI • React / TypeScript • APIs & Microservices at Scale • MongoDB • GenAI & LLM Systems • Batch Orchestration • Evaluation & Observability

Languages & Web: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Node-adjacent service work

GenAI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini | LangChain/LangGraph | RAG | Multi-Agent Systems | Prompt Engineering

Data & Search: PostgreSQL, MongoDB, Redis | FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Kafka | Spark, Databricks

Infra & MLOps: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP | Docker, Kubernetes | Terraform | CI/CD | Batch Orchestration & Retry Logic

PROFESSIONAL EXPERIENCE

AI Engineer - Progress Solutions

Jul 2025 - Present | USA

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift relevance and reduce hallucination.
- Established a structured **evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to surface regressions pre-deployment.

AI Engineer Intern - Vanguard

Jan 2024 - Jan 2025 | USA

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Delivered scalable applications supporting **100,000+ requests daily** across 20+ internal teams, partnering with product, platform, and data engineering.
- Developed and enhanced backend services and AI agents for the internal platform across **25+ agents**, integrating with production data pipelines and observability tooling.
- Engineered content controls for secure, policy-compliant responses, reducing unsafe outputs by **42%** while maintaining quality.

ML & Gen AI Research Assistant - Kent State University

Oct 2023 - Jan 2024 | USA

- Built a privacy-preserving ML pipeline on clinical data (k-anonymity, l-diversity, differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining 99% of baseline predictive performance.
- Designed a composite privacy-utility scoring framework to benchmark six anonymization configurations and identify the optimal operating point.

ML Trainee - Emerson

Jun 2022 - Apr 2023 | India

- Developed supervised regression/classification models for equipment-failure prediction, anomaly detection, and forecasting with end-to-end data preprocessing pipelines.
- Fine-tuned BERT and RoBERTa for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across **29+ sites** - a full-stack product turning a multi-step manual chore into a single inline action.

career-ops

Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications end-to-end without supervision.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0